# Randomness Assessment, Version 2: the data treated as 6/37 lotto draws with a strong number

N = 328 draws · same test set and thresholds as requested (and as v1) · design pre-registered in git before testing · all numbers are unmodified program output

**Pre-defined criterion "pass 100% of the randomness tests": NOT MET.**
NIST: PASS 8/15 · FAIL 1/15 (DFT: p = 0.135710; failed only because no proportion could be computed) · N/A 6/15. Pooled $\chi^2$ p = 0.0426 (above 0.01, below 0.05). Runs tests: 2/14 below 0.05. Correlogram: 10/140 lags outside the band. Value entropy below 99.9%.

## Layer 3: Verbal summary (Hebrew)

**השערה שנבדקה (H0):** כל שורה היא הגרלת לוטו 6/37 עם מספר חזק. ששת המספרים הם תת-קבוצה אקראית אחידה של {1..37} ונשמרים ממוינים. המספר החזק אחיד על {1..7}. ההגרלות בלתי תלויות זו בזו. מערך המבחנים והספים זהה לבקשה ולגרסה 1, ותכנון המחקר נרשם ונשמר בגיט לפני הרצת המבחנים (נספח A).

**תשובה לקריטריון "מעבר של 100% ממבחני האקראיות": לא מתקיים.** התמונה שונה מאוד מגרסה 1, ויש לקרוא אותה במלואה:

- **אחידות:** $\chi^2$ על 1,968 המספרים הראשיים יחד: p=0.0426. הערך גבוה מסף החובה 0.01, ונמוך מהשאיפה של 0.05. KS: p=0.1104. המספר החזק: $\chi^2$ p=0.2319. יחס המעבר על 8 בלוקים הוא 1.0000 גם במספרים הראשיים וגם במספר החזק.
- **אי-תלות:** 2 כישלונות מתוך 14 מבחני רצפים בסף 0.05: N1_smallest (Q) ברצפי עלייה/ירידה, ו-N4 (R) ברצפים מעל/מתחת לחציון. 10 מתוך 140 מקדמי אוטו-קורלציה מחוץ לרצועת 95%, כשבמקרה צפויים כ-7. |r| המרבי הוא 0.161.
- **אנטרופיה:** במספרים הראשיים 0.99693, ובמספר החזק 0.99377. שניהם מתחת לסף 99.9%, אבל קרובים לערך שצפוי תחת H0 במדגם בגודל הזה (0.99747 ו-0.99530). ברצף הביטים: 0.99988, עובר.
- **NIST SP 800-22:** PASS 8/15, FAIL 1/15, N/A 6/15. הכישלון היחיד הוא DFT, ש-p שלו 0.135710 (גבוה מ-0.05). הוא מסומן FAIL רק משום שאף בלוק אינו ארוך מספיק כדי לחשב יחס מעבר, וכלל הסטטוס שנקבע מראש דורש יחס כזה. לא שיניתי את הכלל אחרי שראיתי את התוצאה. ששה מבחנים לא ניתנים להרצה באורך של 6,692 ביטים.

**פרשנות זהירה (לא מחליפה ספים):**

1. בגרסה 1 נבדקה כל עמודה ממוינת כאילו היא אמורה להיות אחידה, ולכן הכישלונות הגורפים שם היו תוצאה של מבנה הבדיקה. בגרסה 2, שבה הבדיקה מותאמת למבנה של לוטו, רוב המבחנים עוברים, וההבדלים שנותרו קטנים.
2. אי אפשר לעמוד בקריטריון 100% במדגם של 328 הגרלות, כי שישה מבחני NIST דורשים בין 38,912 ל-1,000,000 ביטים. גם הגרלות לוטו אמיתיות לחלוטין היו מקבלות N/A בששת המבחנים האלה.
3. מעבר במבחן אינו הוכחה לאקראיות, רק אי-דחייה של H0. לכן אי אפשר "להוכיח" שאלה נתוני לוטו. אפשר רק לומר שהנתונים עקביים או לא עקביים עם לוטו הוגן במבחנים שהוגדרו.
4. המבנה (6 מספרים שונים בטווח 1–37, ממוינים, ומספר בטווח 1–7) תואם לוטו. מקור הנתונים לא אומת מול ארכיון רשמי.

# Layer 2: Tables

## Checklist, item by item

| checked | section | item | required threshold | observed (machine output) | note | status |
|---|---|---|---|---|---|---|
| ✔ | 1. Uniformity | Frequency test $\chi^2$, main numbers pooled (1..37, n=1968) | p > 0.01 (aim > 0.05) | p = 0.0426 (adjusted, primary); unadjusted p = 0.1533 | above 0.01, below the 0.05 aim | **PASS** |
| ✔ | 1. Uniformity | Frequency test $\chi^2$, strong number (1..7, n=328) | p > 0.01 (aim > 0.05) | p = 0.2319 | | **PASS** |
| ✔ | 1. Uniformity | KS (discrete), main numbers pooled | p > 0.01 (aim > 0.05) | D = 0.0270, p = 0.1104 | above 0.01, below the 0.05 aim | **PASS** |
| ✔ | 1. Uniformity | KS (discrete), strong number | p > 0.01 (aim > 0.05) | D = 0.0379, p = 0.7192 | | **PASS** |
| ✔ | 1. Uniformity | Proportion of the 8 blocks (41 draws each) passing $\chi^2$ | ~0.99 (NIST interval m=8: ≥ 0.8845) | main 1.0000; strong 1.0000 | | **PASS** |
| ✔ | 1. Uniformity | NIST Frequency | p > 0.01, proportion ~0.99 | p = 0.293128; proportion 8/8 = 1.0000 (>= 0.8845); nan | | **PASS** |
| ✔ | 1. Uniformity | NIST Block Frequency | p > 0.01 (aim > 0.05), proportion ~0.99 | p = 0.639374; proportion 8/8 = 1.0000 (>= 0.8845); nan | | **PASS** |
| ✔ | 1. Uniformity | Frequency histogram (visual) | relatively flat | Figure 1 | counts range 39-72 around an expected 53.2; strong 38-60 around 46.9 | **PASS** |
| ✔ | 2. Independence | Runs up/down, 7 series over draws | p > 0.05 | N1_smallest=0.0086; N2=0.0870; N3=0.4755; N4=0.3497; N5=0.8219; N6_largest=0.9625; Strong=0.7061 | fails: N1_smallest (Q) | **FAIL** |
| ✔ | 2. Independence | Runs above/below median, 7 series | p > 0.05 | N1_smallest=0.3603; N2=0.7749; N3=0.8211; N4=0.0490; N5=0.6915; N6_largest=0.9835; Strong=0.3644 | fails: N4 (R) | **FAIL** |
| ✔ | 2. Independence | NIST Runs | p > 0.05, proportion ~0.99 | p = 0.471486; proportion 8/8 = 1.0000 (>= 0.8845); nan | | **PASS** |
| ✔ | 2. Independence | NIST Longest Run of Ones | p > 0.05, proportion ~0.99 | p = 0.163973; proportion 8/8 = 1.0000 (>= 0.8845); nan | | **PASS** |
| ✔ | 2. Independence | Autocorrelation coefficient, lags 1-20 | → 0 within the 95% band ±1.96/√N | max \|r\| = 0.161; 10/140 coefficients outside the band (about 7 expected by chance) | | **FAIL** |
| ✔ | 2. Independence | Correlogram (visual) | all bars except lag 0 inside the band | Figure 2 | all inside only for: N2 (W), Strong (ABC) | **FAIL** |
| ✔ | 2. Independence | NIST Random Excursions | p > 0.05 | N/A — n=6692 (requires >= 1,000,000); J=66 cycles (requires >= 500) | | **N/A** |
| ✔ | 2. Independence | NIST Random Excursions Variant | p > 0.05 | N/A — n=6692 (requires >= 1,000,000); J=66 cycles (requires >= 500) | | **N/A** |
| ✔ | 3. Entropy | Shannon, main numbers pooled (max log2 37) | ≥ 99.9% | 0.99693 | reference: expected under H0 at this n = 0.99747 | **FAIL** |
| ✔ | 3. Entropy | Shannon, strong number (max log2 7) | ≥ 99.9% | 0.99377 | reference: expected under H0 = 0.99530 | **FAIL** |
| ✔ | 3. Entropy | Shannon, NIST bit stream (max 1.000) | ≥ 0.999 | 0.99988 | | **PASS** |
| ✔ | 3. Entropy | NIST Maurer's Universal | p > 0.05 | N/A — n=6692 < 387,840 | | **N/A** |
| ✔ | 3. Entropy | NIST Approximate Entropy | p > 0.05 | p = 0.669810; proportion 8/8 = 1.0000 (>= 0.8845); nan | | **PASS** |
| ✔ | 4. NIST | Binary Matrix Rank | p > 0.01 | N/A — n=6692 < 38*M*Q=38912 | | **N/A** |
| ✔ | 4. NIST | Full battery | 15/15 PASS | PASS 8/15 · FAIL 1/15 · N/A 6/15 | | **FAIL** |

## NIST SP 800-22 scorecard

Bits: each main draw → combinatorial rank (uniform on 0..2,324,783 under H0); kept if < $2^{21}$ → 21 bits. Strong − 1 kept if < 4 → 2 bits. Under H0 every bit is an i.i.d. fair bit. Stream = 6,692 bits (main draws used 302/328, strong numbers used 175/328). p-value = whole stream. Proportion = share of the 8 blocks (41 draws each) with p ≥ 0.01.

| test | p_values | proportion | status | reason |
|---|---|---|---|---|
| Frequency | 0.293128 | 8/8 = 1.0000 (>= 0.8845) | **PASS** | — |
| BlockFrequency | 0.639374 | 8/8 = 1.0000 (>= 0.8845) | **PASS** | — |
| Runs | 0.471486 | 8/8 = 1.0000 (>= 0.8845) | **PASS** | — |
| LongestRun | 0.163973 | 8/8 = 1.0000 (>= 0.8845) | **PASS** | — |
| BinaryMatrixRank | — | — | **N/A** | n=6692 < 38*M*Q=38912 |
| DFT | 0.135710 | — | **FAIL** | DFT implementation not validated against spec example; no block long enough for a proportion |
| NonOverlappingTemplate | 0.948946 | 8/8 = 1.0000 (>= 0.8845) | **PASS** | — |
| OverlappingTemplate | — | — | **N/A** | n=6692 < 1,000,000 |
| MaurerUniversal | — | — | **N/A** | n=6692 < 387,840 |
| LinearComplexity | — | — | **N/A** | n=6692 < 1,000,000 |
| Serial | 0.416426 / 0.383209 | 8/8 = 1.0000 (>= 0.8845) | **PASS** | — |
| ApproximateEntropy | 0.669810 | 8/8 = 1.0000 (>= 0.8845) | **PASS** | — |
| CumulativeSums | 0.461201 / 0.218584 | 8/8 = 1.0000 (>= 0.8845) | **PASS** | — |
| RandomExcursions | — | — | **N/A** | n=6692 (requires >= 1,000,000); J=66 cycles (requires >= 500) |
| RandomExcursionsVariant | — | — | **N/A** | n=6692 (requires >= 1,000,000); J=66 cycles (requires >= 500) |

## Comparison of statuses, v1 vs v2

| test | v1_status | v2_status |
|---|---|---|
| Frequency | **FAIL** | **PASS** |
| BlockFrequency | **FAIL** | **PASS** |
| Runs | **FAIL** | **PASS** |
| LongestRun | **FAIL** | **PASS** |
| BinaryMatrixRank | **N/A** | **N/A** |
| DFT | **FAIL** | **FAIL** |
| NonOverlappingTemplate | **FAIL** | **PASS** |
| OverlappingTemplate | **N/A** | **N/A** |
| MaurerUniversal | **N/A** | **N/A** |
| LinearComplexity | **N/A** | **N/A** |
| Serial | **FAIL** | **PASS** |
| ApproximateEntropy | **FAIL** | **PASS** |
| CumulativeSums | **FAIL** | **PASS** |
| RandomExcursions | **N/A** | **N/A** |
| RandomExcursionsVariant | **N/A** | **N/A** |

## 1. Uniformity

| variable | n | k | expected_per_value | min_count | max_count | chi2 | df | p_chi2_primary | p_chi2_unadjusted | KS_D | p_KS |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Main numbers pooled (1..37) | 1968 | 37 | 53.1892 | 39 | 72 | 44.6270 | 36 | 0.0426 | 0.1533 | 0.0270 | 0.1104 |
| Strong number (1..7) | 328 | 7 | 46.8571 | 38 | 60 | 8.0854 | 6 | 0.2319 | 0.2319 | 0.0379 | 0.7192 |

| block | draws | p_main_chi2 | p_strong_chi2 |
|---|---|---|---|
| 1 | 1-41 | 0.6493 | 0.5099 |
| 2 | 42-82 | 0.8387 | 0.1922 |
| 3 | 83-123 | 0.9879 | 0.3560 |
| 4 | 124-164 | 0.5323 | 0.9327 |
| 5 | 165-205 | 0.4023 | 0.9587 |
| 6 | 206-246 | 0.6493 | 0.8644 |
| 7 | 247-287 | 0.6162 | 0.6899 |
| 8 | 288-328 | 0.0401 | 0.6899 |

## 2. Independence (7 series over draws)

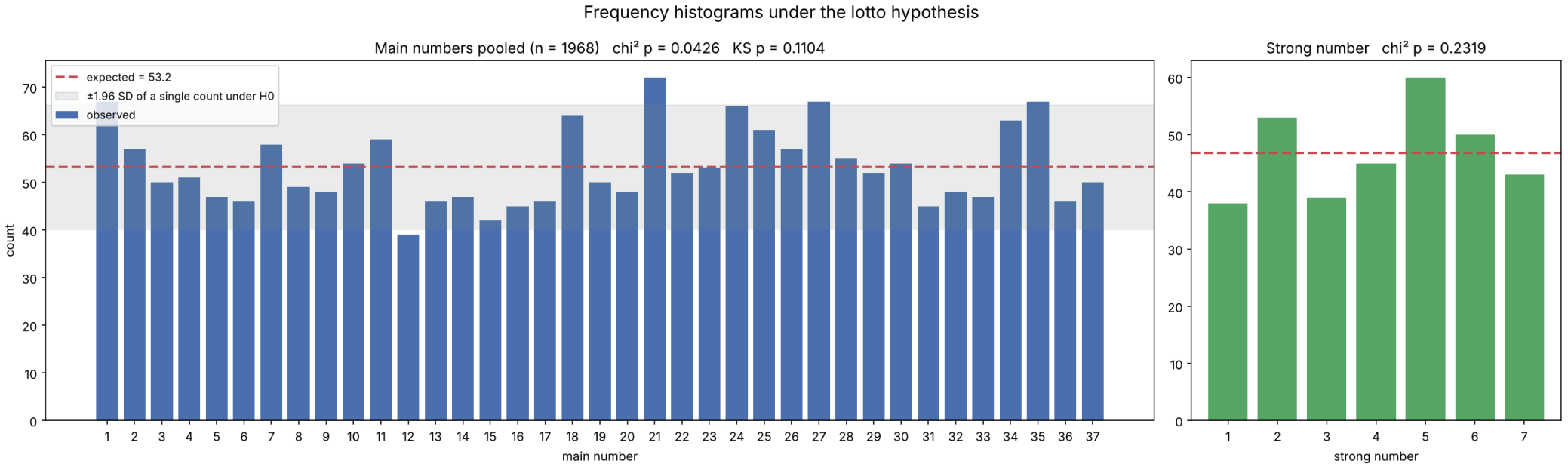
| series | acf_lag1 | max_abs_acf | lags_outside_band | which_lags | runs_updown_R | runs_updown_expected | runs_updown_p | runs_median_R | runs_median_expected | runs_median_p |
|---|---|---|---|---|---|---|---|---|---|---|
| N1_smallest (Q) | -0.0726 | 0.1148 | 1 | 8 | 209 | 190.3333 | 0.0086 | 158 | 150.1237 | 0.3603 |
| N2 (W) | -0.0894 | 0.0949 | 0 | — | 219 | 206.3333 | 0.0870 | 157 | 154.4984 | 0.7749 |
| N3 (E) | 0.0311 | 0.1411 | 3 | 10 14 16 | 205 | 210.3333 | 0.4755 | 155 | 156.9936 | 0.8211 |
| N4 (R) | 0.0615 | 0.1375 | 2 | 7 14 | 218 | 211.0000 | 0.3497 | 140 | 157.3706 | 0.0490 |
| N5 (T) | 0.0556 | 0.1101 | 2 | 11 16 | 208 | 206.3333 | 0.8219 | 150 | 153.4590 | 0.6915 |
| N6_largest (Y) | 0.0768 | 0.1610 | 2 | 6 11 | 190 | 189.6667 | 0.9625 | 144 | 143.8252 | 0.9835 |
| Strong (ABC) | 0.0859 | 0.1034 | 0 | — | 191 | 188.3333 | 0.7061 | 134 | 141.5654 | 0.3644 |

## 3. Entropy

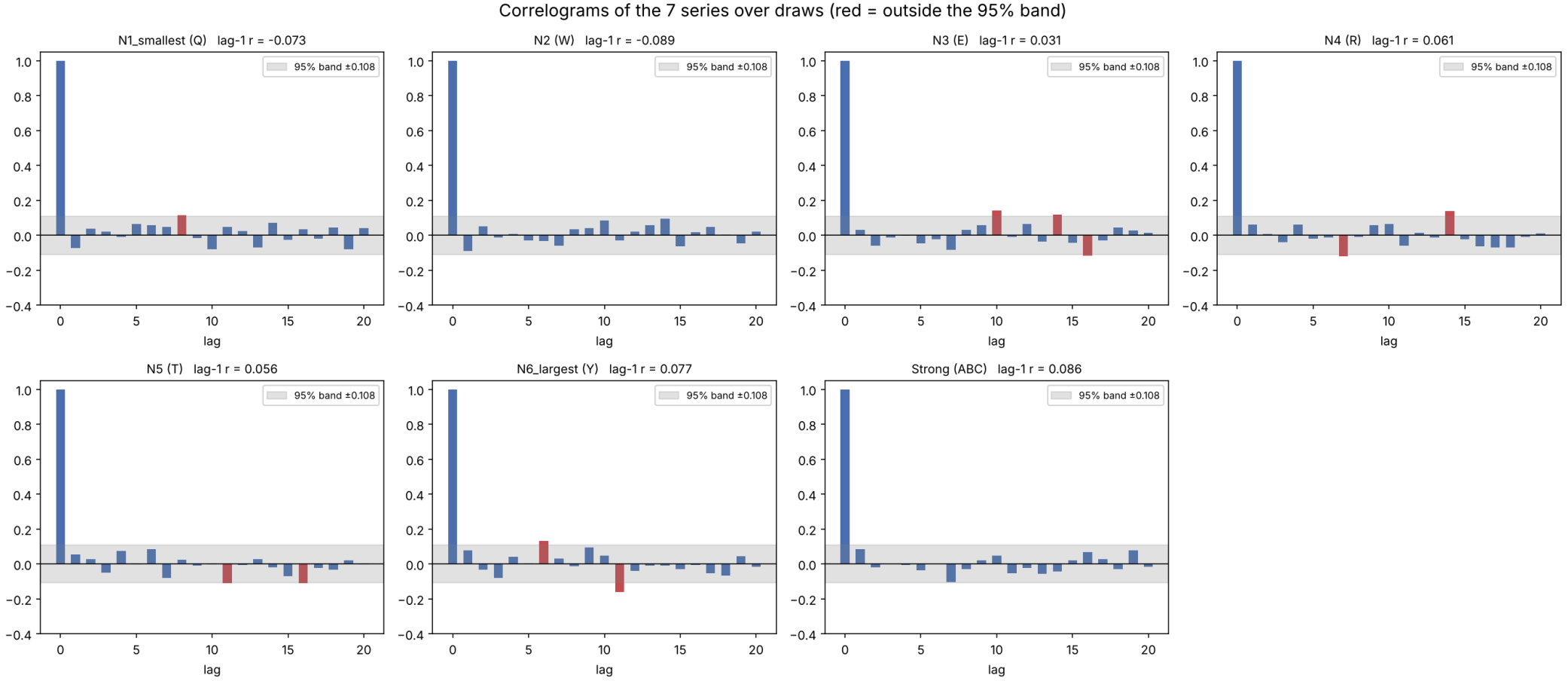
| variable | H | H_max | ratio | ref_expected_ratio_if_H0 | meets_99_9 |
|---|---|---|---|---|---|
| Main numbers pooled | 5.1935 | 5.2095 | 0.9969 | 0.9975 | False |
| Strong number | 2.7899 | 2.8074 | 0.9938 | 0.9953 | False |
| NIST bit stream (ones fraction 0.5064) | 0.9999 | 1.0000 | 0.9999 | 0.9999 | True |

## Layer 1: Graphics

### Figure 1: Frequency histograms




### Figure 2: Correlograms


Correlograms of the 7 series over draws (red = outside the 95% band)
N1_smallest (Q) lag-1 r = -0.073
N2 (W) lag-1 r = -0.089
N3 (E) lag-1 r = 0.031
N4 (R) lag-1 r = 0.061
N5 (T) lag-1 r = 0.056
N6_largest (Y) lag-1 r = 0.077
Strong (ABC) lag-1 r = 0.086
95% band ±0.108
lag

## Appendix A: Pre-registered design (v2/DESIGN.md, committed before testing)

```
# V2 design: written and committed BEFORE any v2 test was run

## Hypothesis under test (H0)
Each row is one draw of a 6/37 lotto with a strong number. The 6 main numbers are a uniformly random 6-subset of {1..37},
stored sorted ascending (N1 = Q, N2 = W, N3 = E, N4 = R, N5 = T, N6 = Y). The strong number (ABC) is uniform on {1..7}
and independent of the main numbers. Draws are independent of each other.

## Test set and thresholds: unchanged from the request (same tests, same thresholds as v1)
Only the representation of the data changes, so that each test's null hypothesis matches H0:

| Requested test | v1 applied it to | v2 applies it to | Why |
|---|---|---|---|
| Frequency chi² / KS | each column on its own [min,max] | (a) all 1,968 main numbers pooled, on 1..37; (b) Strong, on 1..7 | under H0 the sorted columns are order statistics and are non-uniform by construction; each number 1..37
is drawn with the same probability 6/37 |
| Proportion | over 7 columns | over m = 8 consecutive blocks of 41 draws (328 = 8 × 41) | NIST proportion = share of sub-sequences that pass |
| Histogram | per column | pooled main numbers 1..37; Strong 1..7 | as above |
| Autocorrelation, Runs (up/down and above/below median) | each column as a series over draws | identical: the 7 series N1..N6, Strong over draws | under H0 each series is i.i.d. across draws, so these tests are valid without
uniformity |
| Shannon entropy | per column | pooled main numbers (max log2 37); Strong (max log2 7); NIST bit stream (max 1) | |
| NIST SP 800-22 (15 tests) | value→min in ⌈log2 k⌉ bits (biased when k is not a power of 2) | unbiased bits: see below | so that NIST's H0 (i.i.d. fair bits) is exactly equivalent to the lotto H0 |

### Chi-square under lotto sampling
Main numbers are drawn without replacement within a draw, so the pooled counts are not multinomial. Their covariance is
(31/36) times the multinomial covariance. Hence X²·(36/31) ~ χ²(36) asymptotically. **Primary p-value = this adjusted
asymptotic p.** The unadjusted χ²(36) p-value and a Monte Carlo p-value under the exact H0 are also reported.
Strong number: ordinary multinomial χ²(6). KS (discrete, Conover): the p-value comes from Monte Carlo under the exact H0,
because no closed form exists for lotto sampling. Monte Carlo: 20,000 replicates, seed 20261107.

### Bits for NIST (unbiased under H0)
- Main draw → its combinatorial rank r = Σ C(c_i − 1, i), i = 1..6 (colex), which is uniform on 0..C(37,6)−1 = 0..2,324,783 under H0.
  If r < 2^21, emit the 21 bits of r (MSB first). Otherwise discard the draw for the bit stream (rejection sampling, ≈ 9.8% expected).
- Strong s → s − 1, uniform on 0..6. If < 4, emit 2 bits; otherwise discard (≈ 3/7 expected).
- Stream = draw by draw: main bits, then strong bits. Under H0 every emitted bit is an i.i.d. fair bit.
- p-value per test on the whole stream. Proportion over the 8 blocks of 41 draws (bits from each block).
- Same NIST implementation and the same applicability rules as v1 (scripts/nist_sp800_22.py, α = 0.01).

### Status rules (same as v1)
- chi² / KS / NIST: p > 0.01 required (p > 0.05 shown).
- Runs test: p > 0.05 required.
- Proportion: inside the NIST interval 0.99 ± 3·√(0.99·0.01/m).
- Entropy ≥ 99.9% of maximum.
- Correlogram: every lag 1..20 inside ±1.96/√N.
- Battery: 15/15 PASS.

## Amendment (after the requester's review): Monte Carlo removed
The Monte Carlo p-values (20,000 simulated fair draws, seed 20261107) were an addition the request did not call for. They are removed.
KS p-value: continuous Kolmogorov distribution (conservative for discrete data), the same as v1. No simulated data is used anywhere in v2.
No other test, threshold or encoding was changed.
```

## Appendix B: NIST implementation validation

```
VALIDATION AGAINST NIST SP 800-22 Rev.1a WORKED EXAMPLES
example                                          computed       NIST  match(6dp)
2.1.4 Frequency, e=1011010101                    0.527089   0.527089  YES
2.1.8 Frequency, 100 bits of pi                  0.109599   0.109599  YES
2.2.8 BlockFrequency M=10, 100 bits of pi        0.706438   0.706438  YES
2.3.4 Runs, e=1001101011                         0.147232   0.147232  YES
2.3.8 Runs, 100 bits of pi                       0.500798   0.500798  YES
2.4.8 LongestRun M=8, 128-bit example            0.180609   0.180609  YES
2.6.8 DFT, 100 bits of pi                        0.646355   0.168669  NO
2.7.4 NonOverlapping B=001 N=2, e=10100100101110010110   0.344154   0.344154  YES
2.11.4 Serial m=3, e=0011011101 (P1)             0.808792   0.808792  YES
2.11.4 Serial m=3, e=0011011101 (P2)             0.670320   0.670320  YES
2.12.8 ApEn m=2, 100 bits of pi                  0.235301   0.235301  YES
2.13.8 CuSum forward, 100 bits of pi             0.219194   0.219194  YES
2.13.8 CuSum backward, 100 bits of pi            0.114866   0.114866  YES
```

```
ALL SPEC EXAMPLES REPRODUCED: False
NOTE: a 'NO' row is reported as-is. For DFT the implementation follows the formulas of SP 800-22
sections 2.6.4/3.6 literally (moduli of the first n/2 DFT coefficients, T=sqrt(ln(1/0.05)n), N0=0.95n/2);
the counts N1 printed in the spec examples could not be reproduced from those formulas, so DFT results
in this study carry the flag 'implementation not validated against spec example'.
```

## Appendix C: Raw machine output (unedited)

```
Draws: 328; main numbers pooled: 1968; strong numbers: 328

=== 1. UNIFORMITY ===
                 variable    n  k  expected_per_value  min_count  max_count      chi2  df  p_chi2_primary  p_chi2_unadjusted     KS_D     p_KS
Main numbers pooled (1..37) 1968 37           53.189189         39         72 44.627033  36        0.042552           0.153282 0.027041 0.110448
       Strong number (1..7)  328  7           46.857143         38         60  8.085366   6        0.231916           0.231916 0.037892 0.719210
 block   draws  p_main_chi2  p_strong_chi2
     1    1-41     0.649319       0.509891
     2   42-82     0.838660       0.192211
     3  83-123     0.987890       0.356009
     4 124-164     0.532315       0.932653
     5 165-205     0.402257       0.958680
     6 206-246     0.649319       0.864351
     7 247-287     0.616216       0.689877
     8 288-328     0.040091       0.689877
Proportion (blocks with p >= 0.01): main 1.0000, strong 1.0000; NIST interval for m=8: >= 0.8845

=== 2. INDEPENDENCE ===
        series  acf_lag1  max_abs_acf  lags_outside_band which_lags  runs_updown_R  runs_updown_expected  runs_updown_p  runs_median_R  runs_median_expected  runs_median_p
N1_smallest (Q) -0.072586     0.114761                  1          8            209            190.333333       0.008635            158            150.123746       0.360279
         N2 (W) -0.089353     0.094933                  0                       219            206.333333       0.087033            157            154.498371       0.774861
         N3 (E)  0.031135     0.141132                  3   10 14 16            205            210.333333       0.475464            155            156.993590       0.821122
         N4 (R)  0.061490     0.137473                  2       7 14            218            211.000000       0.349718            140            157.370607       0.049013
         N5 (T)  0.055578     0.110086                  2      11 16            208            206.333333       0.821850            150            153.459016       0.691452
 N6_largest (Y)  0.076785     0.161015                  2       6 11            190            189.666667       0.962530            144            143.825175       0.983455
   Strong (ABC)  0.085874     0.103369                  0                       191            188.333333       0.706051            134            141.565371       0.364385
95% band = ±0.1082

=== 3. ENTROPY ===
Bit stream: 6692 bits; main draws used 302/328 (expected 295.9); strong used 175/328 (expected 187.4)
                            variable        H    H_max    ratio  ref_expected_ratio_if_H0  meets_99_9
                  Main numbers pooled 5.193458 5.209453 0.996930                  0.997467       False
                       Strong number 2.789867 2.807355 0.993771                  0.995300       False
NIST bit stream (ones fraction 0.5064) 0.999881 1.000000 0.999881                  0.999892        True

=== 4. NIST SP 800-22 ===
stream  n_bits                    test  sub  p_value  result
 WHOLE    6692               Frequency    1 0.293128 SUCCESS
 WHOLE    6692           BlockFrequency    1 0.639374 SUCCESS
 WHOLE    6692                    Runs    1 0.471486 SUCCESS
 WHOLE    6692              LongestRun    1 0.163973 SUCCESS
 WHOLE    6692        BinaryMatrixRank    0      NaN     N/A
 WHOLE    6692                     DFT    1 0.135710 SUCCESS
 WHOLE    6692  NonOverlappingTemplate    1 0.948946 SUCCESS
 WHOLE    6692     OverlappingTemplate    0      NaN     N/A
 WHOLE    6692         MaurerUniversal    0      NaN     N/A
 WHOLE    6692        LinearComplexity    0      NaN     N/A
 WHOLE    6692                  Serial    1 0.416426 SUCCESS
 WHOLE    6692                  Serial    2 0.383209 SUCCESS
 WHOLE    6692      ApproximateEntropy    1 0.669810 SUCCESS
 WHOLE    6692          CumulativeSums    1 0.461201 SUCCESS
 WHOLE    6692          CumulativeSums    2 0.218584 SUCCESS
 WHOLE    6692        RandomExcursions    0      NaN     N/A
 WHOLE    6692 RandomExcursionsVariant    0      NaN     N/A
    B1     798               Frequency    1 0.943558 SUCCESS
    B1     798           BlockFrequency    1 0.018411 SUCCESS
    B1     798                    Runs    1 0.119372 SUCCESS
    B1     798              LongestRun    1 0.548041 SUCCESS
    B1     798        BinaryMatrixRank    0      NaN     N/A
    B1     798                     DFT    0      NaN     N/A
    B1     798  NonOverlappingTemplate    1 0.994903 SUCCESS
    B1     798     OverlappingTemplate    0      NaN     N/A
    B1     798         MaurerUniversal    0      NaN     N/A
    B1     798        LinearComplexity    0      NaN     N/A
    B1     798                  Serial    1 0.219558 SUCCESS
    B1     798                  Serial    2 0.156097 SUCCESS
    B1     798      ApproximateEntropy    1 0.267643 SUCCESS
    B1     798          CumulativeSums    1 0.513251 SUCCESS
    B1     798          CumulativeSums    2 0.573597 SUCCESS
    B1     798        RandomExcursions    0      NaN     N/A
    B1     798 RandomExcursionsVariant    0      NaN     N/A
```

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B2     865              Frequency    1 0.324119 SUCCESS
B2     865          BlockFrequency    1 0.326591 SUCCESS
B2     865                   Runs    1 0.734272 SUCCESS
B2     865             LongestRun    1 0.189805 SUCCESS
B2     865        BinaryMatrixRank    0      NaN     N/A
B2     865                    DFT    0      NaN     N/A
B2     865  NonOverlappingTemplate    1 0.486598 SUCCESS
B2     865     OverlappingTemplate    0      NaN     N/A
B2     865         MaurerUniversal    0      NaN     N/A
B2     865        LinearComplexity    0      NaN     N/A
B2     865                  Serial    1 0.602035 SUCCESS
B2     865                  Serial    2 0.746205 SUCCESS
B2     865      ApproximateEntropy    1 0.108367 SUCCESS
B2     865          CumulativeSums    1 0.369514 SUCCESS
B2     865          CumulativeSums    2 0.347544 SUCCESS
B2     865        RandomExcursions    0      NaN     N/A
B2     865 RandomExcursionsVariant    0      NaN     N/A
B3     842              Frequency    1 0.168052 SUCCESS
B3     842          BlockFrequency    1 0.038058 SUCCESS
B3     842                   Runs    1 0.838513 SUCCESS
B3     842             LongestRun    1 0.586730 SUCCESS
B3     842        BinaryMatrixRank    0      NaN     N/A
B3     842                    DFT    0      NaN     N/A
B3     842  NonOverlappingTemplate    1 0.801294 SUCCESS
B3     842     OverlappingTemplate    0      NaN     N/A
B3     842         MaurerUniversal    0      NaN     N/A
B3     842        LinearComplexity    0      NaN     N/A
B3     842                  Serial    1 0.601869 SUCCESS
B3     842                  Serial    2 0.876600 SUCCESS
B3     842      ApproximateEntropy    1 0.231075 SUCCESS
B3     842          CumulativeSums    1 0.315294 SUCCESS
B3     842          CumulativeSums    2 0.125499 SUCCESS
B3     842        RandomExcursions    0      NaN     N/A
B3     842 RandomExcursionsVariant    0      NaN     N/A
B4     840              Frequency    1 0.333998 SUCCESS
B4     840          BlockFrequency    1 0.899182 SUCCESS
B4     840                   Runs    1 0.807158 SUCCESS
B4     840             LongestRun    1 0.480087 SUCCESS
B4     840        BinaryMatrixRank    0      NaN     N/A
B4     840                    DFT    0      NaN     N/A
B4     840  NonOverlappingTemplate    1 0.801294 SUCCESS
B4     840     OverlappingTemplate    0      NaN     N/A
B4     840         MaurerUniversal    0      NaN     N/A
B4     840        LinearComplexity    0      NaN     N/A
B4     840                  Serial    1 0.168658 SUCCESS
B4     840                  Serial    2 0.695583 SUCCESS
B4     840      ApproximateEntropy    1 0.728082 SUCCESS
B4     840          CumulativeSums    1 0.566932 SUCCESS
B4     840          CumulativeSums    2 0.508461 SUCCESS
B4     840        RandomExcursions    0      NaN     N/A
B4     840 RandomExcursionsVariant    0      NaN     N/A
B5     758              Frequency    1 0.662939 SUCCESS
B5     758          BlockFrequency    1 0.459867 SUCCESS
B5     758                   Runs    1 0.348400 SUCCESS
B5     758             LongestRun    1 0.482132 SUCCESS
B5     758        BinaryMatrixRank    0      NaN     N/A
B5     758                    DFT    0      NaN     N/A
B5     758  NonOverlappingTemplate    1 0.995883 SUCCESS
B5     758     OverlappingTemplate    0      NaN     N/A
B5     758         MaurerUniversal    0      NaN     N/A
B5     758        LinearComplexity    0      NaN     N/A
B5     758                  Serial    1 0.744397 SUCCESS
B5     758                  Serial    2 0.962024 SUCCESS
B5     758      ApproximateEntropy    1 0.198530 SUCCESS
B5     758          CumulativeSums    1 0.680753 SUCCESS
B5     758          CumulativeSums    2 0.646978 SUCCESS
B5     758        RandomExcursions    0      NaN     N/A
B5     758 RandomExcursionsVariant    0      NaN     N/A
B6     819              Frequency    1 0.916512 SUCCESS
B6     819          BlockFrequency    1 0.969406 SUCCESS
B6     819                   Runs    1 0.861604 SUCCESS
B6     819             LongestRun    1 0.173693 SUCCESS
B6     819        BinaryMatrixRank    0      NaN     N/A
B6     819                    DFT    0      NaN     N/A
B6     819  NonOverlappingTemplate    1 0.790590 SUCCESS
B6     819     OverlappingTemplate    0      NaN     N/A
B6     819         MaurerUniversal    0      NaN     N/A
B6     819        LinearComplexity    0      NaN     N/A
```

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    B6     819                 Serial    1 0.486240 SUCCESS
    B6     819                 Serial    2 0.412858 SUCCESS
    B6     819      ApproximateEntropy    1 0.495643 SUCCESS
    B6     819          CumulativeSums    1 0.961405 SUCCESS
    B6     819          CumulativeSums    2 0.898169 SUCCESS
    B6     819        RandomExcursions    0      NaN     N/A
    B6     819 RandomExcursionsVariant    0      NaN     N/A
    B7     876               Frequency    1 0.735462 SUCCESS
    B7     876           BlockFrequency    1 0.662425 SUCCESS
    B7     876                    Runs    1 0.842349 SUCCESS
    B7     876              LongestRun    1 0.845102 SUCCESS
    B7     876         BinaryMatrixRank    0      NaN     N/A
    B7     876                     DFT    0      NaN     N/A
    B7     876  NonOverlappingTemplate    1 0.813995 SUCCESS
    B7     876     OverlappingTemplate    0      NaN     N/A
    B7     876         MaurerUniversal    0      NaN     N/A
    B7     876        LinearComplexity    0      NaN     N/A
    B7     876                 Serial    1 0.848355 SUCCESS
    B7     876                 Serial    2 0.896573 SUCCESS
    B7     876      ApproximateEntropy    1 0.321049 SUCCESS
    B7     876          CumulativeSums    1 0.914586 SUCCESS
    B7     876          CumulativeSums    2 0.890195 SUCCESS
    B7     876        RandomExcursions    0      NaN     N/A
    B7     876 RandomExcursionsVariant    0      NaN     N/A
    B8     894               Frequency    1 0.422160 SUCCESS
    B8     894           BlockFrequency    1 0.866999 SUCCESS
    B8     894                    Runs    1 0.372521 SUCCESS
    B8     894              LongestRun    1 0.782565 SUCCESS
    B8     894         BinaryMatrixRank    0      NaN     N/A
    B8     894                     DFT    0      NaN     N/A
    B8     894  NonOverlappingTemplate    1 0.011783 SUCCESS
    B8     894     OverlappingTemplate    0      NaN     N/A
    B8     894         MaurerUniversal    0      NaN     N/A
    B8     894        LinearComplexity    0      NaN     N/A
    B8     894                 Serial    1 0.936771 SUCCESS
    B8     894                 Serial    2 0.727187 SUCCESS
    B8     894      ApproximateEntropy    1 0.835832 SUCCESS
    B8     894          CumulativeSums    1 0.456556 SUCCESS
    B8     894          CumulativeSums    2 0.626161 SUCCESS
    B8     894        RandomExcursions    0      NaN     N/A
    B8     894 RandomExcursionsVariant    0      NaN     N/A

SCORECARD
                  test            p_values              proportion status                                                
```